

PAPER<i>251: Eta Carinae Homunculus F</i>U<i>Bi</i>i — DPM Invisibility and LENR Force Hierarchy Discovery

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Series: Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

■# PAPER251: Eta Carinae Homunculus FUBii — DPM Invisibility and LENR Force Hierarchy Discovery

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — EtaCarinaeHomunculusFUBiCalculator (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \text{Bigl}(1 - [SSq] \cdot U_{bi} \backslash, \wedge, F_U(r,t) \backslash \text{Bigr}), \quad \text{quad } [SSq] = 0.57$$

Abstract

Eta Carinae is a hyperluminous stellar system of approximately 120 solar masses, undergoing episodic super-Eddington mass loss. The Great Eruption of ~1843 CE ejected ~10–40 M_{sun} in a bipolar Homunculus nebula that is today one of the brightest infrared sources in the sky. With a magnetic field of B₀ = 10⁴ T — one hundred times stronger than the SN 1006 field — and the same characteristic frequency $\omega = 10^{12}$ rad/s, the Eta Carinae system provides a critical test of the UQFF force hierarchy.

The key **uniquely rare discovery** of this paper is **DPM Invisibility**: despite B₀ being 100× stronger than SN 1006 (B₀ = 10⁵ T), the DPM resonance being 100× larger (1.76 × 10⁵ vs 1.76 × 10³), and the resonance force *Fres being proportionally amplified, the total UQFF buoyancy force FU_Bi remains identical* to SN 1006 at +2.11 × 10²⁸ N. The magnetic field is completely invisible to the final buoyancy result.

This DPM invisibility occurs because $F_{LENR} = k_{LENR} \cdot (\omega_{LENR}/\omega)^2$ is independent of B₀. At $\omega = 10^{12}$, $F_{LENR} \sim 6.17 \times 10^3$ N overwhelms *Fres* by ~3 orders regardless of B₀. The force hierarchy is *LENR > neutron > Newtonian » DPMresonance* in this frequency regime — a fundamental discovery about the structure of UQFF physics.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~7,500	ly	HIPPARCOS/HST
Age (since Great Eruption)	t	5.681 × 10 [?]	s (~180 yr)	1843 CE
Stellar mass	M	120 M _{sun} = 2.387 × 10 ³²	kg	Chandra/JWST model
Homunculus radius	r	6.17 × 10 ¹⁶	m (~20 ly)	HST spatial
X-ray luminosity	L _X	10 ³⁵	W	Chandra 2023

Parameter	Symbol	Value	Units	Source
Magnetic field	B0	10²⁴ T	T	100× SN 1006
System frequency	ω	10 ¹²	rad/s	Same as SN 1006
Eddington factor	M	1.5	—	Super-Eddington

2. Core Physics: DPM Invisibility

2.1 DPM Resonance — 100× SN 1006

$$\begin{aligned} \text{DPM_resonance (Eta Car)} &= 2 \cdot \mu_B \cdot B_0 / (\hbar \cdot \omega) \\ &= 2 \times 9.274\text{e-}24 \times 1\text{e-}4 / (1.0546\text{e-}34 \times 1\text{e-}12) \\ &\sim 1.76 \times 10^5 \text{ [100× SN 1006 value } 1.76 \times 10^3 \text{]} \end{aligned}$$

The resonance force $F_{\text{res}} = 2 \cdot q \cdot B_0 \cdot v \cdot \sin\theta \cdot \text{DPM_resonance}$ is proportional to $B_0 \times \text{DPM_resonance} \propto B_0^2$ — at $B_0 = 10^{24}$ T, F_{res} is 10,000× the SN 1006 value.

2.2 LENR Force — B0-Independent

The LENR force depends only on θ_{LENR} and ω :

$$\begin{aligned} F_{\text{LENR}} &= k_{\text{LENR}} \times (\theta_{\text{LENR}} / \omega)^2 \\ &= k_{\text{LENR}} \times (2\pi \times 1.25 \times 10^{12} / 10^{12})^2 \\ &\sim 6.17 \times 10^3 \text{ N} \end{aligned}$$

There is **no B0 dependence** in F_{LENR} . For both SN 1006 ($B_0 = 10^{25}$) and Eta Carinae ($B_0 = 10^{24}$), F_{LENR} is identical.

2.3 DPM Invisibility Ratio

$$\text{DPM_visibility_ratio} = F_{\text{res}} / F_{\text{LENR}}$$

For SN 1006: $F_{\text{res}} (\text{SN1006}) / 6.17 \times 10^3 \ll 1 \propto \text{DPM invisible}$. For Eta Car: $F_{\text{res}} (\text{EtaCar}) / 6.17 \times 10^3 \ll 1 \propto \text{DPM still invisible, despite } 10,000\times F_{\text{res}} \text{ amplification}$.

The DPM resonance term is submerged under F_{LENR} by at least 33 orders at $\omega = 10^{12}$ for any physically reasonable B_0 .

2.4 Force Hierarchy Theorem

$$\begin{aligned} \text{Force hierarchy at } \omega &= 10^{12}: \\ F_{\text{LENR}} &\sim 6.17 \times 10^3 \text{ N [dominant – } 10^{45} \times \text{Newtonian]} \\ F_{\text{neutron}} &\sim 10^6 \text{ N [knot/coherence stabilisation]} \\ F_{\text{Newt}} &\sim GM/r^2 \cdot |x_2| \sim 0(\text{few}) \text{ N} \\ F_{\text{res}} &\ll F_{\text{LENR}} \text{ [DPM invisible regardless of } B_0\text{]} \\ F_{\text{rel}} &\ll F_{\text{LENR}} \text{ [relativistic sub-dominant]} \end{aligned}$$

2.5 Super-Eddington Luminosity Context

Eta Carinae's super-Eddington luminosity ($M = L/L_{\text{Edd}} \sim 1.5$) drives the 500 AU Homunculus through radiation pressure. The Eddington luminosity:

$$\begin{aligned} L_{\text{Edd}} &= 4\pi G M c / \kappa_{\text{es}} = 4\pi \times 6.674\text{e-}11 \times M_{\text{EtaCar}} \times 2.998\text{e}8 / 0.2 \\ &\sim \text{few} \times 10^{38} \text{ W [for } 120 M_{\text{sun}}\text{]} \end{aligned}$$

The FDE dark energy coupling $k_{DE} \times L_X = 10^{33} \times 10^{35} = 10^5$ N captures the luminosity contribution to FUBi — 3 orders larger than SN 1006's contribution (10^2 N), confirming that even the luminosity difference does not affect the final FUBi when FLENR dominates.

3. DPM Invisibility Theorem

Theorem (UQFF DPM Invisibility at $\omega_0 = 10^{12}$): For any astrophysical system with $\omega_0 = 10^{12}$ rad/s, the DPM magnetic resonance force $F_{res} \propto B_0^2 / \omega_0$ is negligible in the F_{UBi} integral for all physically observed magnetic fields $B_0 = 10^2$ T. The ratio $F_{res}/F_{LENR} = k_{charge} \cdot B_0^2 \cdot v / (k_{LENR} \cdot (\omega_{LENR}/\omega_0)^2)$ is bounded above by $\sim 10^{-28}$ for $B_0 = 10^4$ T, $\omega_0 = 10^{12}$, confirming that **magnetic field strength is invisible to UQFF buoyancy** in this frequency regime.

Corollary: In this regime the UQFF force hierarchy is $LENR > \text{neutron} > \text{Newtonian} \gg \text{DPM} > \text{DE} > \text{relativistic}$. Only LENR and neutron physics materially determine FUBi.

4. Observational Predictions / Validation

DPM invisibility test: UQFF predicts FUBi should be identical for SN 1006 and Eta Carinae despite $100\times$ different B_0 . If future UQFF validation on additional high-B systems confirms this, DPM invisibility is a universal law of the $\omega_0 = 10^{12}$ class.

****Chandra L_X probe:**** $L_X = 10^{35}$ W (Eta Car) vs 10^{32} W (SN 1006) — 3-orders L_X difference. Yet F_{U_Bi} is identical. This confirms $F_{DE} \ll F_{LENR}$ at this ω_0 , providing a direct test of LENR dominance: if Chandra measures any system with $\omega_0 = 10^{12}$ at any luminosity from 10^{31} – 10^{35} W, UQFF predicts the same F_{U_Bi} .

JWST Homunculus 3D: The asymmetric JWST infrared structure of the Homunculus provides f_{TRZ} (triadic resonance zone factor) constraints through the spatial distribution of emitting gas relative to the bipolar axis.

5. References

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